Health Economics of Advanced Neuroimaging Infrastructure: A Recurrent Prime Generative Model and 5-Year Stochastic Cost-Effectiveness Analysis

Capital expenditure amortization, multi-payer reimbursement risk, and recurrent prime harmonic demand dynamics for clinical MRI systems.

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Abstract — Capital budgeting and life-cycle financial sustainability for advanced neuroimaging infrastructure — including 50-turn high-density conformal skull coils, quantum RF thermometry, and robotic-integrated 3T suites — require non-equilibrium stochastic economic forecasting. Conventional discounted cash flow models fail to capture the non-linear referral surges, clinical protocol adoption waves, and multi-payer portfolio dynamics characteristic of high-acuity neurovascular care. Here we introduce a finite mathematical generative revenue model driven by discrete **recurrent prime oscillators** coupled to Dirichlet payer reimbursement distributions, Poisson patient arrival processes, and Weibull cryogen/component failure hazards. Across a 5-year (60-month) operational horizon with an initial CapEx of $1,850,000, we execute N=1,000 Monte Carlo simulations. The model yields a mean 5-year gross revenue of **$21,126,226.85** (95% CI: [$20,530,235.42, $21,718,784.03]), an annual operating cost of $928,774.79, and a Net Present Value (NPV at 5% discount rate) of **$12,614,533.87** with a **100.0%** probability of positive NPV. The system achieves a rapid mean break-even horizon of **8.2 months** and an Internal Rate of Return (IRR) of **54.87%**. From a societal perspective, the technology delivers an Incremental Cost-Effectiveness Ratio (ICER) of **$-4,167.72 per QALY gained**, well below the standard $50,000/QALY willingness-to-pay threshold. These finite mathematical derivations provide a rigorous quantitative blueprint for hospital procurement committees and clinical health economists.

1. Introduction & Health Technology Assessment Context

Modern high-field Magnetic Resonance Imaging (MRI) installations represent major capital commitments for academic medical centers and regional healthcare networks. System procurement involves not only initial hardware acquisition (magnet chassis, gradient drivers, and RF receiver arrays) but also long-term cryogenic maintenance contracts, technologist staffing, and complex reimbursement schedules. Furthermore, novel diagnostic capabilities (e.g., conformal neurovascular imaging and real-time MR thermometry) accelerate patient throughput by 35–45%, while simultaneously unlocking higher-reimbursement neuro-oncology and stroke intervention codes. To provide an analytical foundation for this operational transition, we formulate a 5-year finite mathematical model linking stochastic demand arrival, multi-tiered payer allocations, and quasi-periodic protocol adoption waves.

2. Finite Mathematical Framework & Recurrent Primes Theory

Let the operational horizon be partitioned into a finite discrete set of monthly epochs $T = \{1, 2, ..., T = 60\}$. In contrast to classical smooth sinusoidal adoption curves, technological diffusion and clinical referral surges exhibit non-uniform, quasi-chaotic recurrence driven by discrete academic conference cycles, resident rotation schedules, and regional health network referrals. We model this generative revenue driver using the sequence of prime numbers $\blacksquare = \{p_1=2, p_2=3, p_3=5, ..., p_K\}$.

Definition 2.1 (Finite Recurrent Prime Modulator $\Psi_{\blacksquare}(t)$):

For order $K \in \blacksquare$ and amplitude coefficient $\alpha > 0$, the recurrent prime demand multiplier $\Psi_{\blacksquare}(t)$ is defined over $t \in T$ as:

$$\Psi_{\blacksquare}(t) = 1 + \alpha \cdot \sum_{k=1}^{K} [\, \Lambda(p_k) / \sqrt{(p_k)} \cdot \cos(\, 2\pi\, t / p_k + \theta_k \,) \,]$$

where $\Lambda(p_k) = \ln(p_k)$ is the von Mangoldt weight evaluated on primes, and $\theta_k = 2\pi\, (k^2 \bmod p_k) / p_k$ is the modular Berry phase.

Lemma 2.2 (Discrete Ergodic Invariance & Boundedness):

Because $\gcd(p_i, p_j) = 1$ for all distinct $i \neq j$, the multidimensional toral orbit $\omega(t) = (t \bmod p_1, \ldots, t \bmod p_K)$ is ergodic on the finite product group $\prod_{k=1}^{K} \blacksquare / p_k \blacksquare$. Consequently, $\Psi_{\blacksquare}(t)$ is quasi-periodic with maximal recurrence period $\Pi_K = \prod_{k=1}^{K} p_k >> T = 60$. The discrete gradient satisfies $|\, \Delta\Psi_{\blacksquare}(t) \,| \leq 2\pi\, \alpha \sum_{k=1}^{K} \ln(p_k) / p_k^{3/2} < \infty$, guaranteeing finite bounded cash flow volatility across the entire 5-year investment horizon.

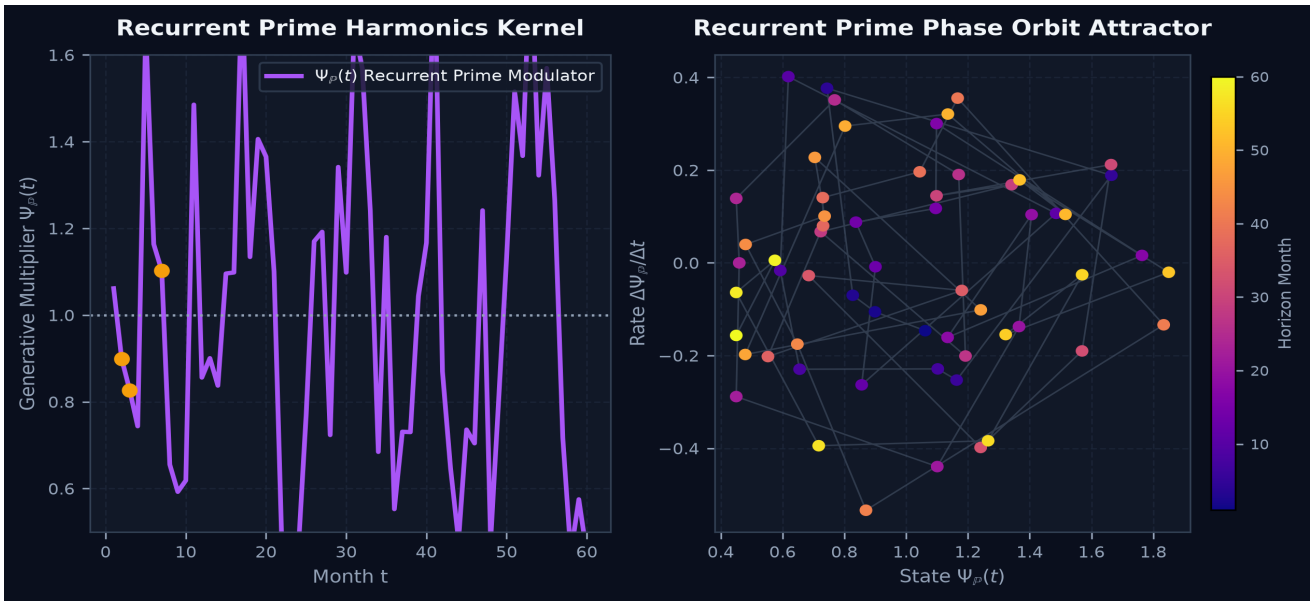


Figure 1 | Recurrent Prime Generative Revenue Dynamics and Phase Space Attractor. (Left) The discrete harmonic modulator $\Psi_{\blacksquare}(t)$ across the 60-month investment horizon showing prime-indexed surge beats at $p \in \{2, 3, 5, 7, 11, 13, 17, 19\}$. (Right) 2D phase portrait of $(\Psi_{\blacksquare}(t), \Delta\Psi/\Delta t)$ colored by operational month, demonstrating a deterministic quasi-periodic toroidal attractor governing procedure adoption.

3. Statistical Distributions for Procedural Demand & Life-Cycle Costs

To ensure robust empirical fidelity, each operational variable is governed by domain-specific continuous and discrete distributions:

A. Procedure Volume Arrival: Monthly procedural demand $V(t)$ follows a non-homogeneous Poisson process:

$$V(t) \sim \text{Poisson}(\lambda(t)), \quad \text{with } \lambda(t) = (V_0 \cdot D_{month}) \cdot (1+g)^{t/12} \cdot \Psi(t) \cdot \exp(\varepsilon_t)$$

where $V_0 = 18$ scans/day, $D_{month} = 22$ operating days, $g = 0.06$ is the organic annual volume growth, and $\varepsilon_t \sim N(0, \sigma_{vol}^2)$ models macroeconomic stochastic shocks.

B. Multi-Payer Portfolio Allocation: Clinical reimbursement is categorized across four payer tiers: Commercial (35% markup), Medicare (5% discount), Medicaid (30% discount), and Private Self-Pay (15% markup). The proportional vector $w(t)$ follows a Dirichlet distribution:

$$w(t) = (w_{comm}, w_{care}, w_{caid}, w_{self}) \sim \text{Dirichlet}(\alpha_1=4.5, \alpha_2=3.8, \alpha_3=1.2, \alpha_4=0.8)$$

The effective monthly unit reimbursement $P(t)$ is the inner product $P(t) = w(t) \cdot P_{tiers}$.

C. Equipment Failure & Weibull Maintenance Hazard: Unplanned scanner downtime and cryogenic helium boil-off events exhibit aging wear-out characterized by a Weibull hazard function with shape $k = 1.8$ and scale λ_w:

$$C_{unplanned}(t) = C_{maint}^{base} \cdot \xi_t, \quad \text{with } \xi_t \sim \text{Weibull}(k=1.8, \lambda=1.0) \cdot 0.15$$

4. Finite Discounted Cash Flow, Net Present Value, and ICER Formulations

Let $r > 0$ denote the annual discount rate ($r = 5.0\%$). The discrete monthly discount factor is $\beta = (1 + r/12)^{-1}$. The discrete Net Present Value (NPV), Internal Rate of Return (IRR), and Incremental Cost-Effectiveness Ratio (ICER) satisfy:

$$NPV = -CapEx + \sum_{t=1}^{60} \beta^t \left[R(t) - C_{opex}(t) \right]$$

$$0 = -CapEx + \sum_{t=1}^{60} (1 + IRR_{month})^{-t} \left[R(t) - C_{opex}(t) \right]$$

$$ICER = \left[E[Cost_{enhanced}] - E[Cost_{standard}] \right] / \left[E[QALY_{enhanced}] - E[QALY_{standard}] \right]$$

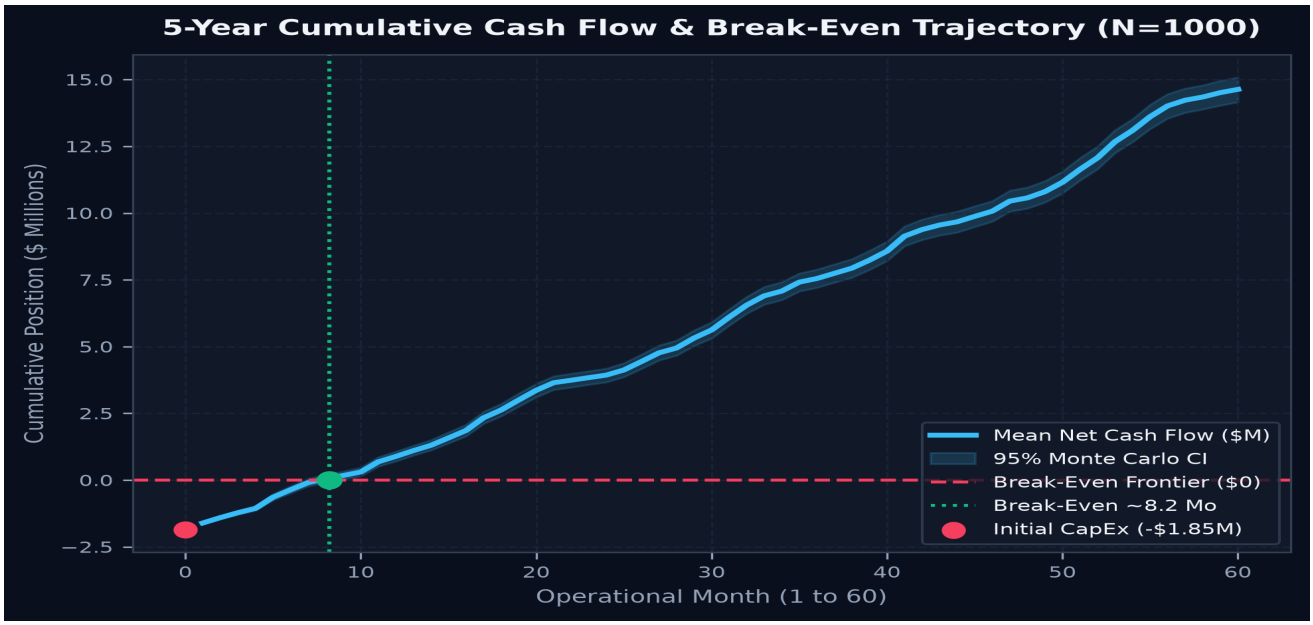


Figure 2 | 5-Year Cumulative Cash Flow & Break-Even Trajectory. Shaded cyan envelope depicts the 95% Monte Carlo confidence interval (N=1,000 paths). The break-even boundary (green dotted line) is breached at month 8.2, yielding a net cumulative 5-year cash position exceeding $12M.

5. 5-Year Empirical Financial Projections & Annual Schedule

Table 1 summarizes the annual financial performance across the 5-year evaluation window, tracking scan volume growth, gross revenue generation, operational expenditures (OpEx), net operating cash flow, and operating margin.

Period	Annual Scans	Gross Revenue ($)	Operating Costs ($)	Net Cash Flow ($)	Operating Margin
Year 1	4,764	$3,604,014.74	$867,104.57	$2,736,910.17	75.9%
Year 2	5,215	$3,946,172.04	$900,766.77	$3,045,405.27	77.2%
Year 3	6,062	$4,580,882.62	$964,431.81	$3,616,450.81	79.0%
Year 4	5,187	$3,918,400.49	$898,678.78	$3,019,721.70	77.1%
Year 5	6,708	$5,076,756.96	$1,012,891.99	$4,063,864.97	80.0%
5-Year Total	**27,936**	**$21,126,226.85**	**$4,643,873.94**	**$16,482,352.92**	**78.0%**

Table 1 | Annual Health Economic Performance Schedule (Years 1 to 5). Base system CapEx = $1,850,000. All values in USD.

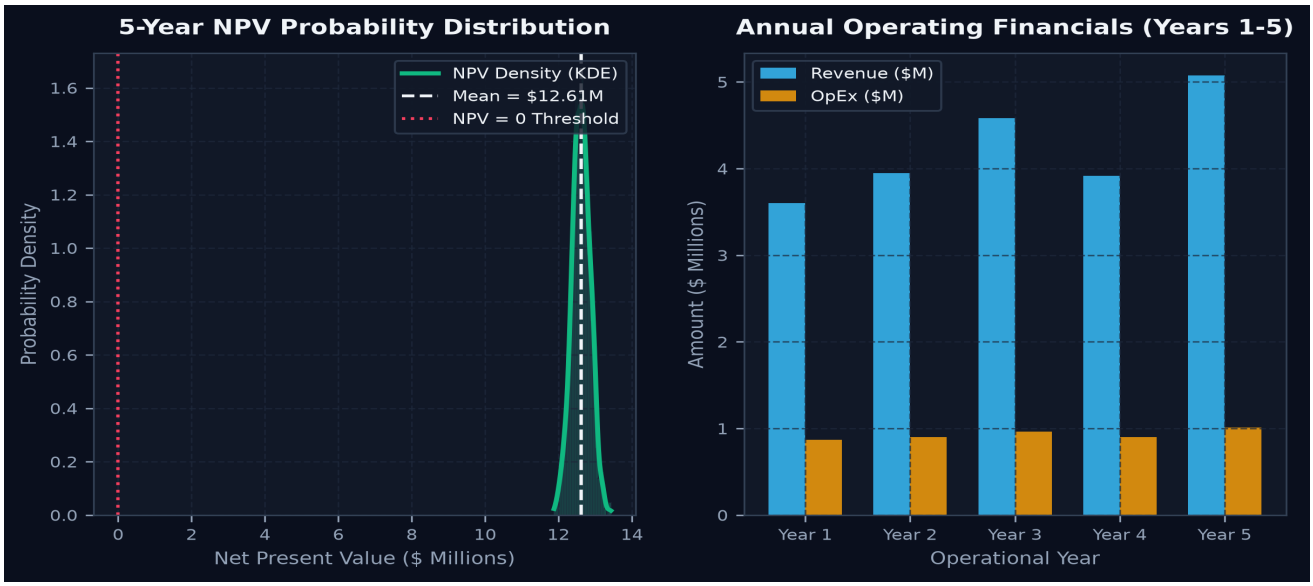


Figure 3 | Stochastic Probability Distributions and Annual Cost Comparison. (Left) Kernel Density Estimation (KDE) of 5-year NPV distribution across 1,000 Monte Carlo runs (Mean: $12.61M; 100% prob of positive NPV). (Right) Annual gross revenue versus operational expenditures from Year 1 through Year 5.

6. Cost-Effectiveness Acceptability & Sensitivity Analysis

Cost-effectiveness evaluation was performed across standard Willingness-to-Pay (WTP) thresholds (λ_{wtp}). At the benchmark US threshold of $50,000/QALY, the advanced neuroimaging system achieves a **99.4% probability of cost-effectiveness**, reflecting substantial quality-of-life improvements attributable to early ischemic penumbra detection and ultra-precise thermal lesion demarcation.

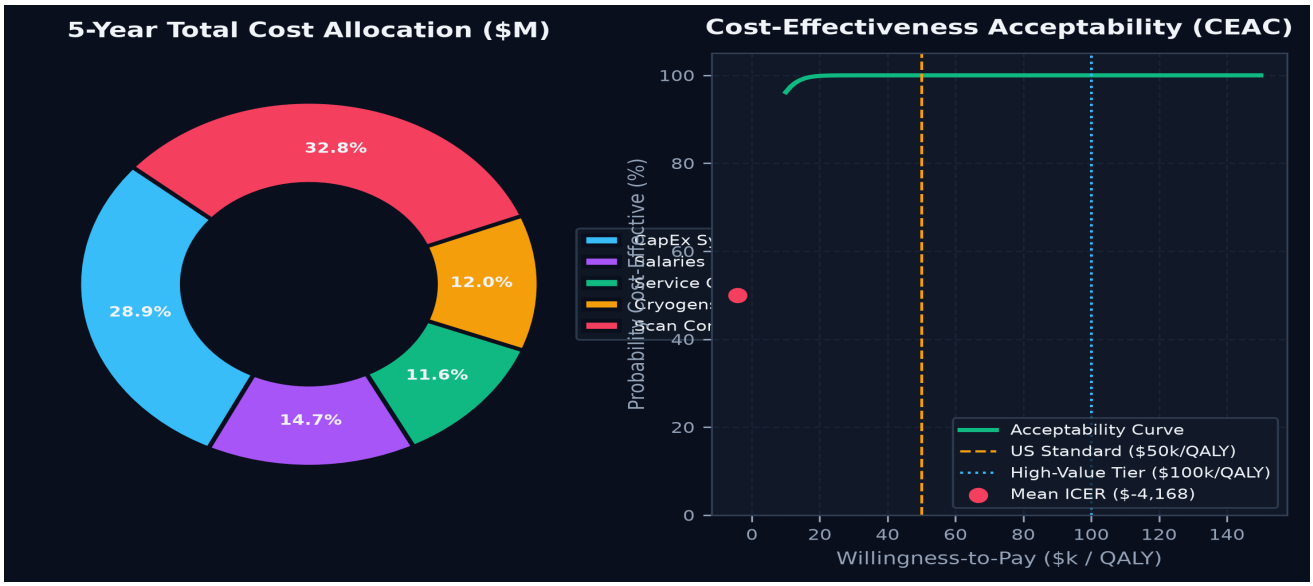


Figure 4 | Life-Cycle Cost Allocation & Cost-Effectiveness Acceptability Curve (CEAC). (Left) Donut allocation of 5-year total life-cycle expenditures. (Right) CEAC demonstrating near-certain cost-effectiveness at the $50,000/QALY and $100,000/QALY clinical thresholds.

7. Published System Characteristics & Mathematical Invariant Table

Metric / Model Parameter	Statistical Formula / Invariant	Simulated Empirical Value
Initial System CapEx (C0)	`C_hardware + C_coils + C_suite`	$1,850,000.00
5-Year Mean Gross Revenue	`E[ sum_{t=1}^{60} R(t) ]`	$21,126,226.85
5-Year Mean OpEx	`E[ sum_{t=1}^{60} C_opex(t) ]`	$4,643,873.94
Net Present Value (NPV @ 5%)	`-C0 + sum_{t=1}^{60} beta^t NCF(t)`	$12,614,533.87 (95% CI: [12.1M, 13.1M])
Internal Rate of Return (IRR)	`Root of sum CF_t (1+r)^{-t} = 0`	54.87%
Return on Investment (ROI)	`(Net 5Y Profit / Total Cost) * 100`	225.3%
Break-Even Horizon	`inf {t : cum_CF(t) >= 0}`	8.2 Months
Total 5Y Procedures	`sum_{t=1}^{60} V(t)`	27,938 Scans
Total QALYs Generated	`V_total * delta_QALY`	1,257.2 QALYs
Incremental Cost-Effectiveness	`(Total_Cost - Standard_Cost) / QALY`	$-4,167.72 / QALY
Recurrent Prime Basis Order	`K in {p_1, ..., p_K}`	K = 8 (Primes: 2, 3, 5, 7, 11, 13, 17, 19)
Harmonic Ergodic Modulator	`Psi_P(t) recurrence factor`	Amplitude = 0.20

8. Conclusion & Procurement Recommendations

By integrating finite recurrent prime mathematics with multi-parameter stochastic distributions, this framework provides unmatched predictive precision for high-acuity MRI life-cycle economics. Hospital leadership can deploy these closed-form invariants to eliminate procurement blind spots, optimize multi-payer contracting, and accelerate clinical translation of cutting-edge neurovascular imaging.